



EXPLANATION

$V4 = \{ I, \sigma_v, C2, \sigma_h \}$

"Thinking examines its own frame."

Informative / Explanatory

Owl Semaphore — Explanation

Informative Companion — Origin Story, Archetype Rationale, Accessibility

Carey James Balboa

Independent DNS Security Researcher

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OWL SEMAPHORE SYSTEM — MASTER PROOF



(Informative companion to the System Specification — v2.0.1)

This document is **informative**, not normative. It tells the origin story, the design rationale, and the audience reasoning that produced the Owl Semaphore. The normative algebra and asset rules live in `OWL-SEMAPHORE-SYSTEM.md` and the four state specifications. Where this document offers warmer wording (especially for METACOGNITIVE), the normative spec retains the scientifically precise version.



1. What the Owl Semaphore is, in one paragraph

The Owl Semaphore is a four-state visual system for marking how a claim, document, dataset, or finding should be evaluated before belief, challenge, or action. The four states — NORMATIVE, NON-NORMATIVE, CRITICAL, METACOGNITIVE — form the Klein four-group V_4 under composition of reflections and 180° rotation. Each state is encoded redundantly through color, orientation, and a textual label so that the state identity remains recoverable when any one channel fails. In a single human sentence: **four owls tell the reader what kind of thinking they are looking at — standard, exploration, inversion, or self-audit.**

1A. Core Framing — Why This Notation Exists at All

The semaphore is a visual notation. Its design target is *compatibility* with established mathematics, accessibility practice, and carefully bounded analogies from epistemology and cognitive science — not a unified theory of everything, and not a claim over the whole of human knowledge.

The four states are not novel inventions; they are a notation system for moves that several mature fields already make. Peer review separates normative from non-normative. CVE assignment flags critical. Methodology audits are metacognitive. The Owl Semaphore's job is to give these long-standing moves a **shared visual language** that survives translation across dashboards, PDFs, slide decks, and security advisories — all with the same four transforms from the Klein four-group.

This means the project's correctness criterion is *compatibility* with the fields it borrows from, not originality. Where the notation contradicts what an experienced psychiatrist, philosopher, statistician, computer scientist, or DNS engineer would already say about an artifact, the notation is wrong — not the field.

1B. How to Read This System — the Four Front-Loaded Stories

The four state specifications each open with a narrative story that bridges human intuition to the mathematical formalism. The deliberate ordering inside each state document is:

1. **Story** — an embodied or historical scene the reader can picture.
2. **Transform** — the V_4 operator and its action on coordinates.
3. **Scientific use** — where this state already operates across fields.
4. **Objections and verification** — the integrity, accessibility, and interpretation rules that protect the mark from misuse.

The four stories follow, each with its state name, the V_4 operator, the story itself, and a one-line claim.



NORMATIVE — *The Proven Ground*

Operator: $I, \det = +1, (x,y) \rightarrow (x,y)$.

Stand upright, the room is the room. Newton's mechanics was operationally validated for roughly two centuries; Einstein's relativity then *extended and contained* it as a limiting case at low velocity and weak gravity, rather than replacing or falsifying it. Within its domain Newtonian mechanics is still normative — and that is exactly what “normative” means here: *operationally validated within a stated domain and scope*, not “true everywhere.”

One-line claim: the work has been done and the foundation holds — within its stated scope, the evidence supports it.

NON-NORMATIVE — *Da Vinci's Wings*

Operator: $\sigma_v, \det = -1, (x,y) \rightarrow (-x,y)$.

Stand in front of a mirror; you are still upright, but left and right have swapped. Leonardo's bird-flight notebooks were a rigorous exploratory record; powered flight came four centuries later, inside the accumulated history of aeronautics.

One-line claim: rigorous exploration that has not finished yet — facing the other direction, without claiming to have replaced the canonical view.

CRITICAL — *The Manhattan Moment*

Operator: $C_2, \det = +1, (x,y) \rightarrow (-x,-y)$.

Do a handstand; both axes flip. Manhattan-Project physicists formally examined whether a nuclear detonation could ignite the atmosphere (LA-602) and ruled it out before action.

One-line claim: your own proof has reversed your position; a vanishingly unlikely catastrophic hypothesis must be resolved, not waved away.

METACOGNITIVE — *The Observer's Mirror*

Operator: $\sigma_h, \det = -1, (x,y) \rightarrow (x,-y)$.

As a *structural analogy only*, Gödel's incompleteness theorems established that a sufficiently powerful consistent formal system cannot prove its own consistency from within itself; we use that shape as an analogy for needing a frame outside the original one, **not** as proof of any claim about cognition, psychology, or color ontology. As a *heuristic illustration* of perceptual frame disruption — not formal psychophysical evidence — a child bends over and looks between their legs and sometimes finds the lost object the upright frame was filtering out.

One-line claim: not a finding about the subject — a finding about the instrument.

The Closing Bridge in §13 explains why these four maneuvers, taken together, are a notation for what every careful field already does.



The full narrative for each state lives in its state document:

- *The Proven Ground* — OWL-1-NORMATIVE.md §1A
 - *Da Vinci's Wings* — OWL-2-NON-NORMATIVE.md §1A
 - *The Manhattan Moment* — OWL-3-CRITICAL.md §1A
 - *The Observer's Mirror* — OWL-4-METACOGNITIVE.md §1A
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2. Where the Owl Semaphore came from — DNS Tool

The Owl Semaphore was not invented to decorate pages. It grew inside the DNS Tool — a long-running research and operations effort that needed to mark, with discipline, the difference between what is observed and what is inferred, between what is asserted and what is audited, and between what is standard and what is exploratory.

DNS Tool already separated several things its operators had to keep separate:

- **observed fact vs. analytical inference** (DNS Tool confidence framework)
- **assertion vs. audit** of standards and resolver behavior
- **success vs. rate-limited vs. error vs. partial** result states for queries
- **standard vs. operational evidence** in the research corpus (DNS Tool corpus)

DNS Tool's "good net citizen" posture also imposed minimal-footprint queries, adaptive rate awareness, documented external data sources, and honest rate-limit reporting. Every part of that posture is an epistemic discipline before it is a network discipline. The Owl Semaphore is the visual-algebra layer that makes that discipline legible to a reader who is not inside the codebase.

In the corpus, the Owl Semaphore had already begun to function as an active classification legend with the lines:

- "This is the standard."
- "This reflects the standard."
- "This inverts the standard."
- (a fourth state describing the audit of the frame itself)

That fourth line — the audit of the frame — is what this release tightens. The earlier wording, "*This audits the standard,*" was insufficient because the audit at this state is **not directed at an external standard**. It is directed at the observer's own evaluative frame. We say so explicitly now: **the observer audits the frame**, or, in plain English, **thinking examines its own frame**.

3. Why four states, not two

Standards culture is comfortable with two states: normative and non-normative. RFC 2119 and RFC 8174 together give a precise vocabulary for the normative/non-normative split, with their BCP 14 meaning attaching to the keywords *when*, *and only when*, *they appear in all capitals*



(RFC 2119; RFC 8174). ISO/IEC Directives Part 2 and W3C Recommendations work in a similar register.

Two states are useful, but they are not enough for the actual congruence problem the DNS Tool faces. A claim, an artifact, or a finding can fail in at least four distinct ways:

1. It can fail to follow the standard it claims to follow. (Need: a baseline reference — **NORMATIVE**.)
2. It can be a legitimate, structured alternative to the standard. (Need: a state for **NON-NORMATIVE** reflection.)
3. It can be deliberately inverted against the baseline for adversarial or falsification analysis. (Need: a state for **CRITICAL** inversion — red-team / falsification.)
4. The observer can be the failure mode — the analyst’s frame of evaluation is what is filtering the truth out of the data. (Need: a state for **METACOGNITIVE** frame-audit — *thinking examines its own frame*.)

The two-state register collapses (3) and (4) into “not normative,” which loses both the adversarial register and the self-audit register. The four-state register keeps them separate so the reader can act differently in each one.

4. Why V_4 — the math

The four states need to *close* under composition: doing any two of them in succession must land you back inside the same set. Otherwise the system would leak into unmarked, unspecified states, and the discipline would erode.

The smallest finite group that fits the four-state behavior we need is the Klein four-group V_4 (Vierergruppe, Wolfram MathWorld; nLab, Klein four-group). In the visual plane it is realized by the symmetry group of a non-square rectangle: the identity, vertical reflection, horizontal reflection, and 180° rotation. It is a **finite subgroup of the orthogonal group $O(2)$ isomorphic to V_4** (equivalently, the dihedral group D_2); it is **not** $O(2)$ itself (Knill, Harvard Math 22b, Unit 8: The orthogonal group).

What V_4 buys us:

- **closure** — composing any two states yields a state already in the set
- **reversibility** — every state is its own inverse ($g^2 = I$)
- **four stable transforms** — no more, no fewer
- **a Cayley table** that fixes how transitions compose, eliminating arbitrary “fifth states”

What V_4 does **not** buy us: it does not guarantee that the four labels carry the right epistemic meaning, and it does not by itself guarantee correctness or security of any artifact tagged with an owl. Group structure is closure, not truth (nLab). The interpretive rules in the state specs do that work.



5. Why the owl

The Owl of Athena carries an old human association with wisdom, vigilance, and nocturnal pattern recognition. Used loosely, that association would be decorative. In the Owl Semaphore it is constrained by math: the owl is the carrier on which the four V_4 transforms act, and the transforms are what give the owl its epistemic meaning.

The owl is, in Peirce's terms, mostly **iconic** (it visually resembles the wise-observer archetype) but partly **symbolic** (the assignment of color and orientation to state is by convention learned by the reader) (SEP, Peirce's theory of signs). The notation uses several of Bertin's visual variables — color hue (selective and associative), orientation (selective), and shape (selective) — paired with a textual label (Moody's dual-coding principle) so that the reader gets multiple, independent cues (Bertin 1967/1983; Moody 2009; Wilke, Fundamentals of Data Visualization). Bertin originally identified **six** retinal variables (size, value, texture, color, orientation, shape) in addition to the two planar position dimensions; motion/dynamics was added by later visualization literature, not by Bertin himself.

That is also why the badge is constrained to a finite set of orientations and a fixed shared geometry. Decorative variation would dissolve the algebra. The owl is the carrier; V_4 is the alphabet.

6. METACOGNITIVE — thinking examines its own frame

This is the state whose explanatory wording matters most, because its English label is the easiest to misread.

Metacognition is the cognitive-science term for monitoring and regulation of one's own cognitive processes — *thinking about thinking* (metacognitive reflection review, PMC 11368986). It is not synonymous with “reflection” or “philosophy.” It is the deliberate inspection of the **machinery of evaluation** itself, not of the object under evaluation.

The σ_h operator in V_4 — the horizontal reflection that flips the vertical axis — is the visual analogue of that move. As a **heuristic illustration of perceptual frame disruption** (not rigorous psychophysical evidence and not a formal grounding for epistemic auditing), consider the **through-the-legs maneuver**: search a room normally, fail to find the target, bend over, and view the same room upside-down through your legs. The room has not changed. You have not changed as an agent. The evaluative frame has been inverted. What was being filtered out by your upright-priors can sometimes become visible. That illustration is meant to make the operational meaning of the METACOGNITIVE state memorable, not to prove it.

This release adopts the canonical phrasing:

- **Scientific / normative voice:** “*The observer audits the frame.*” (used in OWL-SEMAPHORE-SYSTEM.md, OWL-4-METACOGNITIVE.md, the four-state ledger on every generated PDF, the README state table)



- **Warmer explanatory voice:** “*Thinking examines its own frame.*” (used in this explanation, in teaching materials, in introductory bridges)

The earlier line “*This audits the standard*” is **deprecated** as of v2.0.0. It conflated *frame-audit* with *standard-audit* and read as if the auditor and the audited were both external — the opposite of what the state encodes. Two things are worth keeping clear:

- **Metacognitive prompts surface bias; they do not eliminate it.** The cognitive-science literature is explicit on this point (Flavell 1979 / metacognition review, PMC 11368986); the Owl Semaphore must not claim more than that. ICD 203 does *not* use the word *metacognition*, but its analytic-tradecraft requirements — methods that reveal and mitigate the impact of assumptions and cognitive biases — functionally parallel metacognitive self-monitoring (ICD 203, §B).
- **METACOGNITIVE is observer-level, not object-level.** A CRITICAL owl inverts the *object’s* assumptions; a METACOGNITIVE owl inverts the *observer’s* frame. They are different operators in V_4 and they encode different work.

7. Accessibility — color is not the only carrier

Color matters in this system: gold for NORMATIVE, teal for NON-NORMATIVE, red for CRITICAL, amethyst for METACOGNITIVE. But color cannot do the work alone.

- About **8% of males and 0.5% of females of Northern-European descent** have red-green color vision deficiency; rates vary by population (PMC global review, 12385717).
- **WCAG 2.2 Success Criterion 1.4.1 (Use of Color)** prohibits color from being the only visual means of conveying information (W3C SC 1.4.1).
- **Section 508 §302.3** requires at least one visual mode of operation that does not require user perception of color (section508.gov).
- **CRITICAL** in particular is intentionally low-contrast: a red owl on a warm-red field. If color were the only carrier, CRITICAL would be unrecoverable for red-axis (protan) CVD users. So it cannot be the only carrier.

The Owl Semaphore therefore requires that **every state identity be recoverable from three independent channels**:

1. **color** — the assigned palette
2. **orientation** — the V_4 transform (upright/inverted $\times$ right/left-facing) applied to the canonical owl
3. **textual label and context** — the literal state token (NORMATIVE, NON-NORMATIVE, CRITICAL, METACOGNITIVE) printed alongside the badge, together with the math/quote tuple

This is the project’s “color is not the only carrier” rule. It is design-level conformance with WCAG SC 1.4.1; full empirical Level-AA testing (automated checkers, CVD simulation, user testing — see Wilke, *Fundamentals of Data Visualization*) is scoped to a future release.



8. Who the Owl Semaphore is for — audience archetypes (ordered entry doors)

DNS Tool’s hardest design problem was not technical. It was that several expert cultures had to read the same artifacts without alienating each other. The Owl Semaphore is one of the answers: a notation that lets each archetype find what it needs without flattening the others.

The ordering below is intentional: it moves from the most skeptical readers who want proof artifacts in hand, through the readers who want discipline-of-method, through the readers who want correctness against an external register, and finally to the reader who only wants a memorable shape. Every archetype is told *which entry door* to use — story, transform, scientific use, or objections-and-verification — so they can argue the PDF from the layer that matches their training without being forced through someone else’s.

1. **Hackers and operators** — entry door: *objections and verification*. They want proof and reproducibility; they read the integrity manifest, the SHA-3-512 hashes, and the banner-tuple test. The four stories are background colour for them, not the load-bearing argument.
2. **OSINT and intelligence analysts** — entry door: *scientific use and citation discipline*. They want sourcing discipline; they read the citations and confidence language. ICD 203 analytic tradecraft is treated as a reference framework, not as a binding external requirement (ICD 203; ICD 208). The Manhattan-Project example in *The Manhattan Moment* is in their idiom: “examine the catastrophic hypothesis, even if remote, before acting.”
3. **DNS engineers and protocol implementers** — entry door: *transform*. They want protocol correctness; they read the standards and the rules of engagement. The V_4 Cayley table is the part they will argue.
4. **Standards / RFC readers** — entry door: *transform plus objections*. They want normative discipline; they read the BCP 14 keyword usage and the deprecation note for “*This audits the standard.*” The stories are illustrative, not normative.
5. **Cybersecurity and incident responders** — entry door: *story* → *scientific use*. The *Manhattan Moment* story names *0-day* explicitly as the cybersecurity instantiation of CRITICAL; they will argue the C_2 owl from CVE workflow before they argue it from group theory (CISA, “Known Exploited Vulnerabilities Catalog”; NIST, *Zero-day attack* glossary).
6. **Psychiatrists, psychologists, and cognitive scientists** — entry door: *story* → *scientific use*. *The Manhattan Moment* names *state incongruence / frame-discrepant finding* (older clinical literature used *ego-dystonic*, which is a descriptor and not a current formal diagnostic category, and not equivalent to CRITICAL); *The Observer’s Mirror* names *metacognition* and uses Gödelian self-reference as a *structural analogy* only. These readers argue from clinical and cognitive grounds first; the V_4 algebra is a notation overlay, not a claim about cognition itself (Flavell 1979 / metacognition review, PMC 11368986).
7. **Philosophers and formal logicians** — entry door: *story* → *transform*. *The Observer’s Mirror* uses Gödel’s incompleteness theorems as a structural analogy (not as a grounding or proof of the system); *The Manhattan Moment* references *aporia* in the Platonic / productive-perplexity sense, not as a synonym for CRITICAL. These readers argue the algebra against its



own consistency before they argue its applications (SEP, “Gödel’s Incompleteness Theorems”; SEP, “Aporia” via Socratic method).

8. **Roboticians, sensor-fusion and machine-vision practitioners** — entry door: *scientific use*. *The Observer’s Mirror* states the σ_h posture in their idiom: an instrument reports the world *through a coordinate frame*, and frame audits are a routine part of calibration discipline (ROS REP 105; Thrun, Burgard & Fox, *Probabilistic Robotics*).
9. **Data scientists and visualization practitioners** — entry door: *transform plus accessibility*. They want explicit visual variables and accessible colour; they read Bertin’s retinal variables, Moody’s dual-coding, and WCAG SC 1.4.1.
10. **Historians of science and engineering** — entry door: *story*. *The Proven Ground* and *Da Vinci’s Wings* are written carefully to avoid the most common overclaiming patterns (no direct lineage from Leonardo to Kitty Hawk; no assertion that Newtonian mechanics was “wrong” before relativity).
11. **Serious operators and incident managers** — entry door: *objections-and-verification*. They want explicit rules for when to stop, escalate, or audit; they read the interpretation doctrine.
12. **Public readers and onboarding audiences** — entry door: *story* → *four-owl ledger*. They want a memorable shape and a clear story; the four stories are written for them first.

The four-state algebra was designed to be the smallest set that lets each of these archetypes act differently without breaking the others’ rules.

9. How this connects to the larger Intellectual Resistance framework

The Owl Semaphore sits inside a fourfold operating field that DNS Tool uses to keep work congruent:

- **Verification Principle** — verify the claim
- **Carrier Color Theory** — clean the carrier (don’t let the medium leak meaning the message did not authorize)
- **Symbiotic Net Citizen Principle** — protect the system
- **Versioning and Provenance Rule** — preserve the path

The Owl Semaphore marks the **state of thought before action**. It is the visible state layer that helps the system stay congruent across claim, carrier, code, consequence, and correction. Where a claim’s verification fails, a CRITICAL owl makes that visible. Where an analyst has just inverted their frame to look again, a METACOGNITIVE owl makes that visible. The semaphore does not assert truth; it asserts **what kind of thinking the reader is looking at**.

10. What the Owl Semaphore is not

- **Not a guarantee of truth**. It encodes evaluative position, not correctness.



- **Not a security proof.** Group structure is closure, not security. seL4 demonstrates that critical software can be subjected to machine-checked formal proof (seL4 SOSP 2009; seL4 whitepaper); the Owl Semaphore aspires to analogous discipline (explicit invariants, bounded state spaces, machine-checkable transitions) without claiming the same proof depth.
- **Not an accessibility certification.** It targets WCAG 2.2 Level A intent for SC 1.4.1; empirical Level-AA conformance testing is future work.
- **Not a moral judgment.** A CRITICAL owl does not call the underlying claim wrong; it says someone has subjected it to adversarial analysis.

11. Bridges to DNS Tool and Zenodo

- DNS Tool Owl Semaphore page: <https://dnstool.it-help.tech/owl-semaphore>
- DNS Tool confidence framework: <https://dnstool.it-help.tech/confidence>
- DNS Tool corpus (where Owl Semaphore functions as the live legend): <https://dnstool.it-help.tech/corpus>
- DNS Tool publications: <https://dnstool.it-help.tech/publications>
- Concept DOI (all versions): <https://doi.org/10.5281/zenodo.19473697>
- v2.0.1 version DOI: <https://doi.org/10.5281/zenodo.20419874>
- Previously published v2.0.0 version DOI: <https://doi.org/10.5281/zenodo.20418539>
- Earlier published v1.2.0 version DOI: <https://doi.org/10.5281/zenodo.19474599>

12. Closing

The Owl Semaphore tells the reader what kind of thinking they are looking at. Four owls, four V_4 transforms, four colors, four orientations, four labels — and one rule that ties the whole thing together: *the system encodes position, not truth*. When the METACOGNITIVE owl is present, the system has just done something the system most needs to be able to do: **think about its own thinking**.

13. Closing Bridge — Why This Matters

Why does this matter? Several mature fields already operate with moves that look like these four states — they just lack a shared visual language. Peer review separates normative from non-normative. CVEs flag critical. Methodology audits are metacognitive. The Owl Semaphore offers those long-standing moves a notation system that works on a dashboard, a PDF, a slide deck, and a security advisory — all with the same four transforms from the Klein four-group.

The four front-loaded stories — *The Proven Ground*, *Da Vinci's Wings*, *The Manhattan Moment*, and *The Observer's Mirror* — are deliberately drawn from physics, art-and-engineering history,



nuclear safety, and formal logic respectively. That spread is not decorative. It is the operational claim of the project: the notation must remain *compatible* with established mathematics, accessibility practice, and carefully bounded analogies from epistemology and cognitive science. It does not claim to encompass “all human knowledge”; it claims to be visibly consistent with the fields it borrows from.

When it does, the payoff is small and concrete: a reader who has never seen this system before can look at a marked artifact and immediately know whether they are reading a baseline, an exploration, an inversion, or a frame-audit — and a reader who has spent thirty years in one of the contributing fields can still argue every line on the page from their own training without being told to set their expertise aside.



OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v2.0.1)



NORMATIVE

$T = I \quad \det = +1$

$(x, y) \rightarrow (x, y)$

“This is the standard.”

RFC 2119 MUST / SHALL



NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

$(x, y) \rightarrow (-x, y)$

“This reflects the standard.”

Informative / Advisory (NOTE)



CRITICAL

$T = C_2 \quad \det = +1$

$(x, y) \rightarrow (-x, -y)$

“This inverts the standard.”

RFC 2119 MUST NOT / SHALL NOT



METACOGNITIVE

$T = \sigma_h \quad \det = -1$

$(x, y) \rightarrow (x, -y)$

“The observer audits the frame.”

Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).